

Does distribution make any difference?

Quantitative comparison of collocated and globally distributed projects

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Abstract—Research on globally distributed software development (GSD) assumes the characteristics of the GSD projects to differ from collocated projects. Results of mainly qualitative studies describe communication, coordination, and issues related to team dynamics as typical problems in GSD projects. However, there is a lack of quantitative studies using real project data, and proven research instruments and methods in which collocated and distributed projects are compared systematically. The results of our quantitative analysis show significant differences between these project types concerning team dynamics, teamwork quality, individual satisfaction and project performance. The results are consistently “favoring” the collocated projects. Hypothetically, applying the key principles of Scrum and XP would help to mitigate the problems of GSD projects.

Keywords—teamwork; trust; commitment; identification; performance; globally distributed software development

I. INTRODUCTION

A distinguished body of research within SE science has developed around the issues that the globalization of software development has raised especially during the recent decades [1]. Industrial experience reports and qualitative studies tell a story of communication and coordination problems in projects where the development teams comprise of employees from different countries, time-zones and cultural backgrounds, with scarce possibilities to face-to-face collaboration, getting acquainted with each other, and team-building. Disciplined use of processes and frequent visits have been suggested as good practices [2] [3] in globally distributed development projects (GSD) which is mainly motivated by the promise for increased project efficiency by utilizing the lower-cost pool of software professionals available globally in countries like India, China, and Eastern Europe [4].

GSD research has been highlighting the role of communication and trust [5] [6] [7] in software projects. In addition, increasing attention have been drawn also to the other team dynamical and social factors that may well be negatively affected by the limited physical presence that the distributed development teams need to cope with [8]. As the sum of the various factors that increase the level of complexity of GSD projects, it is also a believed that they are more prone to failure than collocated projects.

However, taking in account the topicality of GSD issues, there are extremely few quantitative analyses comparing globally distributed and collocated industrial software projects and rigorously discussing on the issues that differentiate GSD projects from collocated projects. The GSD research has been mainly descriptive and explorative, and relied on qualitative research methods such as case studies [1].

In this paper we present tentative results from a study, where collocated and globally distributed projects were compared by measuring the levels team dynamics, teamwork quality, and individual satisfaction among the employees in the projects. Employees in distributed and collocated projects were also compared on the grounds of the characteristics of their work and their views on the performance level of the project they worked in.

Our paper sharpens the understanding of the differences between the collocated and distributed software development projects. This understanding is fundamental for all GSD research. By using the established quantitative survey instruments that have been developed in social and organizational sciences, method that is underutilized in GSD research, our paper has also strong methodological contribution.

II. RESEARCH QUESTION

The purpose of this paper is to find preliminary answers to the following question.

Is there a difference between the collocated projects and the distributed projects compared by the following factors?

- Work design
- Team dynamics
- Teamwork quality
- Project performance
- Individual satisfaction

We will analyze the data further and in a near future publish more detailed results in a journal paper. Thus this paper can be viewed as a description of the “research in progress”.

III. RELATED LITERATURE AND THE RESEARCH VARIABLES

In this section we introduce the research instruments and variables we used in the analysis. All the used variables, respective amount of items and results of scale reliability analysis are listed in Table III.

A. Work Design

The rationale for measuring the characteristics of work in GSD projects is based on the way how the global employee resources are utilized in software projects. At least in part of the appeal of global distribution in software development is based on strategies, in which the employee resources responsible of the tasks that are by their nature more routine, repetitive, and requiring relatively lower level of competence and work experience, are sought from countries where pool of software professionals with sufficient education and competences is available with significantly lower costs than in Western Europe and North America. Thus, from this perspective is it reasonable to ask, if (and what kind of) differences there is between the employees at onsite and offsite with respect to their roles, responsibilities, and the qualitative nature of work in software development projects, and what kind of consequences these possible differences may have. For example, Durnell Cramton and Hind's [9] model of the subgroup dynamics in internationally distributed teams suggest that clear and overlapping differences in roles, responsibilities and status between the subgroups may easily cause ethnocentric attitudes and a conflict between the workgroups.

To measure the work characteristics of the employees in the analyzed projects, we used the Work Design Questionnaire (WDQ) developed by Morgeson and Humphrey [10]. WDQ was developed to integrate the key facets from the large amount of earlier measures for characteristics of work, and it provides a general measure for work design with high validity (see the original article). WDQ is an instrument consisting of total of 21 sum variables in four categories (task characteristics, knowledge characteristics, social characteristics, work context). Each sum variable consists of 3-6 items. In our data collection 11 of these 21 variables were represented. However, we modified these variables by in most cases using only one or two items for each variable.

B. Team Dynamics

Trust. Measures of team dynamics included measures for team trust, project commitment, and team identification. Trust is repeatedly named as crucial factor for successful collaboration in distributed software teams. Certain level of trust between people is needed to enable collaboration (sharing of work related and personal information, sharing of responsibilities) between people. In distributed projects efficient collaboration is expected between people who usually don't have much common personal or professional history, which is seen as foundation for development of trust (as well as distrust). Corporate strategies that rely on distribution and outsourcing at high volume may cause anxiety among the employees who are concerned about the changes that the distribution may cause to their work both at short and longer term [5].

There are various definitions for trust. One of the most used [11] defines it as confidence that the trustee won't use received resources opportunistically in a way which could cause harm for the trustor. The measures of trust comprise of items that respectively focus on the different aspects of trust, like benevolence, integrity, competences and predictability (see [12] for review of the features and suitability of the different trust measures). In this study the team trust was measured by using the 16-item scale by Spreitzer and Mishra [13].

Project commitment. Project commitment can be defined as the "sense of duty that the team feels to achieve the project's goals and willingness to do what's needed to make the project successful" [14]. Knowledge intensive occupations relying in individual skills and problem solving abilities within interdependent teams, like software development, a degree of self-direction and adoption of the general goals of the project and motivation to help the goal achievement is needed from the employees. Observed low project commitment should be understood as a warning sign of some problems that may cause a threat for the project success and outcomes. Project commitment was measured with a measure developed by Hoegl, Weinkauff, and Gemuenden [15].

Organizational identification means emotional attachment with the organization and feeling that part of personal identity, the "self", is based on the organizational membership. Instead of the whole company, in project-organizations the most important target for identification can be the project team that works towards the shared goal. Team identification is seen as having positive influence on collaboration and team performance via more efficient communication, coordination, mutual support, and pro-social behavior. In distributed projects the level of identification and attachment with the project team may be especially important because of the limited opportunities to relationship building via face-to-face contacts. In this paper we measured the team identification with scale developed by Mael and Ashforth [16], modified by [17]. According to meta-analysis by [18], the scale by Mael and Ashforth is considered as preferable option to measure organizational identification. In our survey the scale was worded to measure the employees' level of identification with the *project team*.

C. Teamwork Quality and Project Performance

Software development is fundamentally team oriented action. Software development projects employ groups of professionals, who may have special roles based on their expertise like UI design, architecture design, or testing. However, the duties of the team members are highly connected, and tackling the various complexities of software development requires efficient communication in the team and coordination of tasks [19]. To measure teamwork quality, we used the *communication, coordination and mutual support* facets of the teamwork quality measure developed by [20].

Hoegl and Gemuenden [20] also developed a measure for assessing the project performance, which builds on the previous conceptualizations by e.g. [21] and [22]. They separate the team performance and the personal success of the employees, the former comprising of *effectiveness* and

efficiency facets and the latter of *work satisfaction* and *learning* facets. To measure project performance, we used the *effectiveness* and *learning* facets, added with self-designed *resourcing* facet targeted to measure the satisfaction to the amount and quality of employee resources in the project.

D. Individual Satisfaction

To assess employees' fulfillment at individual level, our questionnaire contained measures for *job satisfaction*, *individual team satisfaction*, and *turnover interests*. High turnover in GSD projects has been stated as a problem especially in rapidly developing global IT sourcing hubs found from example in India, China and Eastern Europe. It has been suggested that there may be a connection between low commitment and high turnover in GSD projects [23].

We measured job satisfaction with two different measures. The measure by [24] measures the general job satisfaction with three questions. The Minnesota Satisfaction Questionnaire focuses on different aspects of work satisfaction, like for example personal recognition, advancement possibilities and feeling of worthwhile achievements. A measure by [21] was used to measure individual satisfaction with working as part of the project team. To measure turnover intention a measure based on by Bozeman and Perrew [25] was used.

IV. DATA COLLECTION

The research data were collected between 9/2009 and 6/2010. Software executives of Finnish ICT companies were informed about our data collection project and invited to take part in case that suitable ongoing projects could be identified. Convenient sampling was used to get contact to potential companies. Our aim was to get data both from collocated and distributed projects. Criteria for identifying suitable project were the following:

- The project is a software development project (not a continuous action, i.e. continuous maintenance project)
- During the data collection, the project is at least in the middle in its lifecycle (not a project that has just recently started)
- Based on the amount of personnel working in it, the size of the project should be at least moderate (minimum of 5 employees)
- The project personnel should be fairly dedicated to the project in question

In case that a company showed interest in taking part to data collection and that suitable project(s) was identified, the project and development team management of the given project were contacted and informed about our data collection. The project and team management was encouraged to inform also the development team members about data collection and discuss about their willingness to take part. Procedures of guaranteeing the confidentiality and anonymity were clarified, and the voluntariness of participation was emphasized for the employees. As an incentive to take part, written feedback reports of the basic results and possibility of comparison with the other participating projects were promised. However, the

identity of the other participating companies and projects weren't revealed either during the data collection or in the feedback reports.

As a result, total of 10 software development project teams from 5 Finnish companies took part to data collection. Four of the projects were collocated, meaning that in these projects all members of the development team worked from the same physical location. If any of the development team members were permanently located to some other country, the project was considered as internationally distributed. This was the case with six of the projects. The research data is described in Table I.

TABLE I. RESEARCH DATA DESCRIPTION

Project	Firm	Description	Employees at onsite and offsite	Answer Rate (%)
1	Alfa	Project for porting multiple applications to a new operating environment	3 + 7	90%
2	Alfa	Business reporting software	6	100%
3	Beta	Embedded software platform project	4 + 1 + 1	100 %
4	Beta	Embedded software product	5 + 7	75%
5	Beta	Data visualization software product	4 + 3	86%
6	Zeta	CAD / CAE software product	4	100%
7	Zeta	CAD / CAE software product	5	80%
8	Theta	UI upgrade for existing business management software	2 + 3	100%
9	Theta	Centralized printing solution	7 + 3	80%
10	Epsilon	Electronic document management products	9	100%
			Total 49 + 25	Total 89%

The employees of the projects were contacted personally with electronic survey tool. The survey form comprised of personal background and demographic questions and various established measures focusing on different aspects of work design, group dynamics, teamwork quality, project performance, and individual satisfaction. It took approximately from 15 to 25 minutes to fill the questionnaire. Probably because of the employees own motivation to take part in to this study, the answer rates were exceptionally high. Total answer rate was 89%, and from 5 projects we got answers from all employees.

The questionnaire consisted of background variables and research variables. The background variables are listed in table II. These background variables were included to get insight on the level of demographic homogeneity of the compared groups and its possible influences on the other results. However, a fine

grained analysis of the demographic differences between the the compared groups is not included in this paper.

The table III lists the used research variables, number of items in each variable and respective results of scale reliability analysis for variables comprising of more than 1 item. Chronbach's Alfa was used as the method for scale reliability analysis. The scale reliability analysis show that the scales for measuring aspects of Work design proved to have unacceptable internal reliability, except the Specialization variable (Alfa= .736). However, the scales for all the other research variables can be considered reliable, with Alfa values ranging from .700 to as high as .950. Only exception having unacceptable reliability is the Resourcing variable within the Project Performance, which comprised of two self-designed questions.

TABLE II. BACKGROUND VARIABLES

Role in the Project
Gender
Education level
Work Experience
Official Employer

TABLE III. RESEARCH VARIABLES AND SCALE RELIABILITY

Concept / Instrument	Source	Items	Alfa
Work Design	Morgeson, F.P. and Humphrey, S.E. (2006)		
- Scheduling Autonomy		1	na
- Methods Autonomy		1	na
- Task Variety		2	.637
- Task Significance		2	.494
- Task Identity		2	.543
- Job Complexity		2	.392
- Information Processing		1	na
- Problem Solving		2	.285
- Specialization		2	.736
- Social Support		4	.592
- Feedback from Others		1	na
Team Trust	Spreizer and Mishra (1999)	16	.944
Project Commitment	Hoegl M, Weinkauff K, and Gemuenden H.G. (2004)	5	.884
Team Identification	Mael, F.A., & Ashforth, B.E. (1992)	5	.763
Teamwork Quality	Hoegl, M. and Gemuenden, K. (2001)		
- Communication		7	.724
- Coordination		3	.700
- Mutual Support		5	.818
Team Satisfaction	Gladstein D. (1984)	3	.871
Job Satisfaction (general)	Cammann, C., Fichman, M., Jenkins, D. & Klesh, J. (1979)	3	.871
Job Satisfaction (MSQ)		12	.880
Turnover Intentions	Bozeman D.P., Perrew P. L. (2001)	3	.781
Project Performance	Hoegl, M. and Gemuenden, K. (2001)		
- Effectiveness (Quality)		10	.923
- (Resourcing) *		2	.341
- Learning		4	.708

*Not included in the Project Performance measure by (Hoegl and Gemuenden 2001).

V. RESULTS

A. Background variables

To discover the similarities and differences among the groups being compared in this paper, in this section I present the results of comparing the background variables between the employees working in the collocated and distributed projects.

The table IV shows the role distribution among the employees in collocated and distributed projects. The distribution shows that no striking differences were found by comparing the role division in distributed and collocated projects. Within both project types over 60% of the employees worked as developers.

TABLE IV. ROLE DISTRIBUTION

Roles	Collocated Projects (N=23)	Distributed Projects (N=43)
Project Managers	13,0 %	9,1 %
Developers	69,6 %	61,4 %
Testers	21,7 %	15,9 %
UI Designers	13,0 %	0,0 %
Architects	17,4 %	11,4 %
RE Analysts	4,3 %	6,8 %
Scrum Masters	4,3 %	9,1 %
Product Owners	8,7 %	2,3 %
Team Leaders	4,3 %	0,0 %

Table V shows the gender distribution in collocated and distributed projects. Within both project types over 80% of the employees were males.

TABLE V. GENDER DISTRIBUTION

Gender	Collocated Projects (N=23)	Distributed Projects (N=43)
Female	17,4 %	11,4 %
Male	82,6 %	88,6 %

Table VI shows the education level of the employees in collocated and distributed projects. The share of employees with Master's level degree is remarkably higher in collocated projects. In distributed projects the Bachelor level degree is slightly more common than in collocated projects.

TABLE VI. EDUCATION LEVEL

Education level	Collocated Projects (N=23)	Distributed Projects (N=43)
High School	4,3 %	2,0 %
College / Vocational Education	0,0 %	7,0 %
University level studies, no degree	13,0 %	18,6 %
Bachelor's Degree	30,4 %	37,2 %
Master's Degree	52,1 %	32,5 %
Licentiate or Doctoral Degree	0,0 %	2,0 %

Table VII describes the work experience among the respondents of this study. While employees in collocated and distributed project both had slightly over 5 of work experience with their current (at time of the data collection) employer, the employees in distributed projects had on average 3,5 years more general work experience in software engineering than the employees in collocated projects.

TABLE VII. WORK EXPERIENCE (MEAN)

Work Experience (Years)	Collocated Projects (N=21)	Distributed Projects (N=43)
In Software Engineering	7,2	10,7
With the Current Employer	5,6	5,3

Table VIII shows who was the official employer of the employees in collocated and distributed projects. All of the employees working in the collocated projects had a work contract directly with the company which owned the product being developed or the project. In distributed projects, over 40 % of employees represented some external resourcing service provider, offering professional employee resources for software engineering industry.

TABLE VIII. OFFICIAL EMPLOYER

Employer	Collocated Projects (N=23)	Distributed Projects (N=43)
Self-employed / freelancer	0,0 %	2,3 %
Other	0,0 %	4,6 %
External resourcing service provider	0,0 %	41,8 %
Company owning the product / project	100,0 %	51,1 %

As a conclusion of the comparison considering the background variables, it seems that the two groups consisting of the employees working in collocated and distributed projects are fairly homogenous. There are some differences, mainly in education levels and employer –employee affiliations, but in general comparing these two groups should be free of severe bias based on background variables.

B. Research variables

Table IX presents the results of the T-test, which is a method for revealing significant ($p < .05$) differences of the mean values in measured variables between the groups. In this case, the compared groups were the employees working in the collocated projects and the employees working in the distributed projects. In the table IX, the amount of answers (N), mean value of the answers (M), and standard deviation of the answers (SD) are listed for all the research variables and separated based on the Project type (Collocated / Distributed). The last column (Sig.) presents the level of statistical significance of the differences between the mean values.

TABLE IX. T-TEST RESULTS

	Project type	N	M	SD	Sig.
Work Design	Scheduling	Coll. 23	5,43	0,90	,006*
	Autonomy	Distr. 30	4,60	1,16	
	Methods	Coll. 23	5,78	0,95	,082
	Autonomy	Distr. 43	5,21	1,39	
	Task Variety	Coll. 23	6,04	0,62	,226
		Distr. 42	5,82	0,74	
	Task Significance	Coll. 23	4,96	1,38	,762
		Distr. 42	4,86	1,19	
	Task Identity	Coll. 23	5,22	0,99	,609
		Distr. 42	5,06	1,28	
	Job Complexity	Coll. 23	5,15	1,21	,722
		Distr. 33	5,26	0,99	
	Information Processing	Coll. 23	5,74	1,14	,722
		Distr. 43	5,84	1,02	
Team Dynamics	Problem Solving**	Coll. 22	5,68	0,52	,227
		Distr. 42	5,45	0,98	
	Specialization	Coll. 23	5,74	1,01	,978
		Distr. 43	5,73	0,90	
	Social Support	Coll. 22	5,40	0,80	,530
		Distr. 41	5,27	0,76	
	Feedback from Others	Coll. 23	5,30	1,15	,132
		Distr. 43	4,77	1,46	
	Team Trust	Coll. 21	5,80	0,69	,021*
		Distr. 35	5,32	0,75	
	Project Commitment	Coll. 22	5,91	0,82	,001*
		Distr. 34	5,06	0,95	
	Team Identification	Coll. 20	5,96	0,56	,000*
		Distr. 37	4,31	0,92	
Teamwork Quality	Communication	Coll. 21	5,72	0,58	,001*
		Distr. 38	5,00	0,81	
	Coordination	Coll. 22	5,48	0,97	,002*
		Distr. 41	4,60	1,10	
	Mutual Support	Coll. 22	5,93	0,65	,001*
		Distr. 42	5,22	0,80	
Project Performance	Effectiveness (Quality)	Coll. 15	5,37	0,78	,003*
		Distr. 38	4,43	1,05	
	Resourcing	Coll. 22	5,59	0,93	,002*
		Distr. 42	4,73	1,03	
	Learning	Coll. 23	5,45	0,93	,832
		Distr. 42	5,40	0,80	
Individual Satisfaction	Job Satisfaction (general)	Coll. 23	6,09	0,95	,009*
		Distr. 43	5,41	0,99	
	Job Satisfaction (MSQ)	Coll. 22	5,20	0,64	,019*
		Distr. 37	4,64	0,98	
	Team Satisfaction	Coll. 23	6,04	0,89	,018*
		Distr. 43	5,45	0,98	
	Turnover Intentions	Coll. 23	2,22	1,15	,010*
		Distr. 40	3,13	1,39	

* $p < .05$; **Equal variances not assumed

Work design. Within the Work Design variables only one significant difference were found. The employees in collocated project reported having higher scheduling autonomy than the employees in distributed projects.

Team Dynamics. Within the Group Dynamics variables significant differences between the employees in collocated and distributed projects were found with every variable; *team trust*, *project commitment*, and *team identification*. Within all of these variables the scores of employees in collocated projects were higher and thus more favorable than the scores of employees in distributed projects.

Teamwork Quality. In all of the Teamwork Quality variables, *communication*, *coordination*, and *mutual support*, significant differences were found between employees in collocated and distributed projects. Again the scores showed significantly higher teamwork quality in the collocated projects.

Individual Satisfaction. Scores in Individual Satisfaction variables measuring *job satisfaction*, *team satisfaction*, and *turnover intentions* were all significantly more favorable among the employees in the collocated projects. Both measures of job satisfaction showed similar results.

Project Performance. Of the Project Performance variables, the scores of the *effectiveness* variable measuring the general quality of the project work and output showed that the collocated projects were evaluated performing significantly better than the distributed projects. The *resourcing* variable focused on the sufficient amount and quality of employee resources, and revealed significantly higher satisfaction on employee resources in the collocated projects. No differences in *learning* aspect of project performance were found between collocated and distributed projects.

VI. DISCUSSION

In general, our results showed remarkable differences between collocated and distributed software development projects. With only few exceptions, all the variables related to group dynamics, teamwork quality, project performance, and individual satisfaction significantly differed between these two types of projects. With all the variables showing significant differences, the mean values were more “favorable” for the collocated projects.

Our results strengthen the view of internationally distributed software development projects as more complex than collocated projects with respect to employee relations and collaboration dynamics, as well as project performance. The results of lower jobs satisfaction and higher turnover intentions in the distributed projects can be seen as reflecting the other results. In all the level of consistency that the results showed within group dynamics, teamwork quality, individual satisfaction, and project performance were however somewhat surprising.

In our data some differences in the background variables may well be related to some of the other results. It is not a surprise that in distributed projects part of the employees were officially employed by some external resourcing service

provider company. However, it is reasonable to believe that this share being as high as 41.8 %, it may well have an influence more directly to the development of team identity, project commitment, team trust and teamwork quality, and more indirectly also to the project performance. It seems that employees in distributed projects had a bit longer work history in SE than the employees in collocated projects. However, the education level of employees was a bit higher in collocated projects.

VII. AGILITY AND GSD

While a section of GSD research and literature has concentrated practically on finding out ways how to cope with the phenomena that seem to be at least partially inbuilt in to GSD fundamentals, like issues in intercultural communication and collaboration, and coordination of work between distributed team members (see systematic literature review by [1]), there has been less discussion in general on the relation between team-centric paradigm of software development and global distribution trend. A key question for that discussion would be, *is it possible to maintain the team-centric approach to software development in globally distributed context?* However, insights to that discussion can certainly be adopted from the discussion related to the rise of the “agile methods” and their suitability for GSD projects. The feasibility of using agile methods also in GSD context has been discussed for example by [27] and [28].

The core emphasis of the well-known agile methods, Scrum and XP, is much on the internal dynamics of the development team and the principles how the team should work together. These principles include factors like self-direction, collective ownership, pair work, knowledge sharing and open communication, iterative planning and progress, reliance on individual competences and high motivation (see e.g. [29]). A way to overcome the seemingly fundamental incompatibility between the team-oriented paradigm of software development that have established mostly during the history of collocated development practices, and global distribution, may be found by rigorous application of agile methods like Scrum and XP in GSD projects. There is some evidence that by applying agile methods like Scrum and XP it is possible to increase the project performance via their inbuilt emphasis on the social factors [30]. Our claim is that GSD projects especially would gain much benefits of this effect, as their core problems and challenges seem to be to a great extent social and communicational by their nature.

In contrast with some earlier development process models, the principles of e.g. XP and Scrum contain concrete assumptions about the aspired work conditions and collaboration “atmosphere” within the development team. We also believe that the positive effect of the work towards reaching these conditions in GSD teams would have a positive effect also on the factors related to team dynamics, individual satisfaction, and project performance that we measured in this study. This is a question that should be studied empirically in future research.

VIII. LIMITATIONS

In this paper we, on purpose, haven't been discussing at all about the absolute levels of the research variables among the employees. Despite of the clear *differences* between the employees in collocated and distributed teams, the absolute levels of the variables necessarily aren't alarming. This issue can be considered as the main limitation of our study.

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